SKYGLASS

DIY BUILD GUIDE

Bill of materials

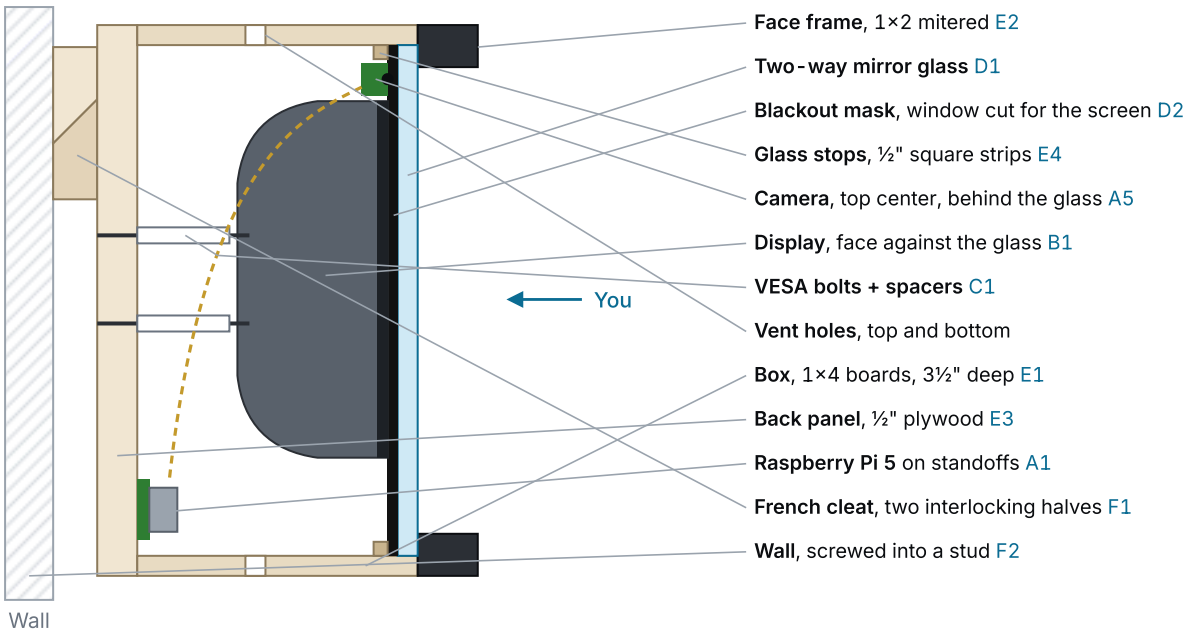
Everything you need to buy to build a SkyGlass mirror around your software license: computer, camera, display, mirror glass, frame, wall mounting and power.

PARTS COST	BUILD TIME	SKILLS	EDITION
$490–$1,050	A weekend	Basic woodworking	October 2026
about $645 for the recommended 24" build	including paint drying	no soldering, no wiring	prices in USD before tax

1 · BEFORE YOU BUY

What you're building

A SkyGlass mirror is a shallow wooden box. The front is two-way mirror glass. Behind it, a computer monitor sits face-down against the glass, with a Raspberry Pi and a small camera alongside. Where the screen is lit, its light shows through the glass; everywhere else you see an ordinary mirror. The box hangs on a French cleat and runs from one wall outlet.



Side section of the recommended build. Not to scale. Codes like D1 refer to the parts lists in section 2.

Decide four things first

01
Display size

Everything else is sized from the display. 24" is the sweet spot: easy to read from across a room and light enough to hang. Most builds use 22–27".

02
Landscape or portrait

Landscape suits a hallway or desk. Portrait (display turned 90°) makes a taller mirror for a dressing area. The parts are the same; only the glass and frame dimensions change.

03
Glass or acrylic

Glass gives a true, flat reflection and is the recommendation. Two-way acrylic is cheaper and lighter but bends the reflection on larger sizes. See the comparison on page 6.

04
Where it hangs, and how power gets there

The finished 24" mirror weighs about 30–35 lb. Plan to screw into a wall stud, and decide whether the power cord shows, runs in a cord cover, or goes through the wall.

Measure your display before you order glass. Record the body width and height (stand removed), the depth at its thickest point, the size of the lit screen area, the VESA hole spacing, and where the HDMI and power ports are. Glass is cut to order and usually can't be returned.

1 · BEFORE YOU BUY

Sizing

Order glass at least 1" larger than the display body on every side, so the frame lip covers the glass edge and the bezel stays hidden. The sizes below are good starting points. Larger glass is fine; the extra area is plain mirror.

DISPLAY	TYPICAL BODY (W × H)	GLASS, LANDSCAPE	GLASS, PORTRAIT	GLASS WEIGHT, 4 MM / 6 MM	FINISHED WEIGHT
21.5–22"	19½ × 11½"	22 × 16" 2.4 sq ft	16 × 26" 2.9 sq ft	5 / 8 lb	≈ 25 lb
23.8–24" (recommended)	21¼ × 12¾"	24 × 18" 3.0 sq ft	18 × 30" 3.75 sq ft	6 / 10 lb	≈ 30–35 lb
27"	24¼ × 14⅜"	28 × 20" 3.9 sq ft	20 × 34" 4.7 sq ft	8 / 13 lb	≈ 40–45 lb

Typical bodies vary by model; use your own measurements. Finished weight includes display, frame, electronics and 6 mm glass.

The three builds priced in this guide

	BUDGET	RECOMMENDED	PREMIUM
Display	22" IPS monitor	24" IPS monitor	27" bright IPS monitor
Mirror	Two-way acrylic, 22 × 16"	Two-way glass, 24 × 18"	Smart-mirror glass, 28 × 20"
Frame	Pine box, painted	Poplar box, painted or stained	Poplar box, painted or stained
Mounting and power	Plywood cleat, visible cord	Aluminum cleat, paintable cord cover	Aluminum cleat, in-wall power kit
Parts total	**≈ $490**	**≈ $645**	**≈ $1,050**

You don't need a keyboard, mouse or spare monitor. Setup (WiFi, license, location, airport and face registration) happens in the SkyGlass phone app. You will need a phone (iPhone or Android), home WiFi with internet, your SkyGlass license key, and a computer with a microSD card slot: it flashes stock Raspberry Pi OS with Raspberry Pi Imager, then runs the SkyGlass install script on the Pi over SSH, as described in the SkyGlass hardware guide.

About prices

Prices are typical US retail in October 2026, before tax and shipping, rounded to the dollar. Raspberry Pi boards went up several times in 2025–26 because of memory costs, so check the current price. Glass prices vary most by region: get two quotes.

2 · PARTS LIST

Computer and camera

SkyGlass runs on a Raspberry Pi 5 with 4 GB of memory. 8 GB adds nothing for SkyGlass, and boards with less than 4 GB struggle with face recognition and the display at the same time.

REF	ITEM	QTY	EST.	NEED
A1	**Raspberry Pi 5, 4 GB** The computer. A Pi 4 (4 GB) also works if you already own one, with slower face recognition. It then needs a regular 15 W USB-C supply and the standard camera cable. Raspberry Pi approved resellers (raspberrypi.com/products/raspberry-pi-5)	1	$110	Required
A2	**Raspberry Pi 27 W USB-C power supply** Use the official 27 W (5.1 V / 5 A) supply. Phone chargers cause brown-outs and random restarts.	1	$12	Required
A3	**Raspberry Pi 5 Active Cooler** Clip-on heatsink and fan. The Pi runs continuously in a closed box, so it needs the fan.	1	$5	Required
A4	**microSD card, 64 GB, A2 rated** 32 GB is the minimum. Buy a name brand (Raspberry Pi, Samsung, SanDisk). An "endurance" card lasts longer under constant use.	1	$15	Required
A5	**Raspberry Pi Camera Module 3 (standard lens)** Small enough to hide behind the glass. Get the standard 75° lens: the wide version makes faces too small at arm's length. NoIR isn't needed.	1	$25	Pick A5 or A5b
A6	**Raspberry Pi 5 camera cable, 500 mm** The Pi 5 uses a smaller 22-pin camera socket, and the cable in the camera box doesn't fit it. 500 mm reaches from the Pi at the bottom of the box to the camera at the top.	1	$3	With A5
A5b	**USB webcam, 1080p, UVC (instead of A5 + A6)** Any small webcam that works without drivers (UVC). It's larger and harder to hide, but it can sit outside the glass, which helps in dim rooms. Pick one with a flat body under 1¼" tall.	1	$30	Pick A5 or A5b
A7	**Micro-HDMI to HDMI cable, 1 m (3 ft)** The Pi 4 and 5 have **micro**-HDMI ports. A right-angle or slim cable is easier to route in a shallow box.	1	$8	Required
A8	**M2.5 nylon standoff kit** Mounts the Pi to the back panel with a gap for airflow. A Pi 5 case fastened with A8's screws or Dual Lock (C2) also works.	1 kit	$8	Required
A9	**Raspberry Pi 5 RTC battery** Keeps the clock right through a power cut when the internet is also down. Plugs into the Pi's battery connector.	1	$5	Optional
A10	**USB microSD card reader** Only if your computer has no SD slot. Used once, to flash Raspberry Pi OS with Raspberry Pi Imager.	1	$8	If needed
	Subtotal, required parts with camera A5		**$186**	

Display

Any ordinary computer monitor works, as long as it meets the requirements below. Check the spec sheet for each one before you buy.

REF	ITEM	QTY	EST.	NEED
B1	**24" IPS monitor, 1920 × 1080, HDMI** Must have: IPS panel (stays readable at an angle) · brightness 250 nits or more (300+ for a bright room; the glass absorbs most of the light) · HDMI input · VESA 100 × 100 mounting holes and a removable stand · a flat screen, not curved · turns back on by itself after a power cut · goes to sleep when the HDMI signal stops. Nice to have: thin, flat back · power and menu buttons on the back or bottom edge · external power brick (less heat inside the box) · power light that can be switched off. Office monitors from Dell, HP, ASUS, Acer and LG. Budget build: 22" (≈ $85). Premium build: 27" at 300+ nits (≈ $160).	1	$110	Required
	Subtotal		**$110**	

Depth check: the recommended box leaves 3¼" behind the glass. If your monitor is deeper than 3" at its thickest point, build the box from 1×6 instead of 1×4 (E1).

Inside the box

Mounting hardware and the one power strip that everything plugs into.

REF	ITEM	QTY	EST.	NEED
C1	**M4 bolts, 50 mm, with nylon spacers and washers** Bolt the monitor's VESA holes to the back panel. Choose spacers so the screen rests lightly against the mask without pushing on the glass. Check your monitor's thread first: most use M4.	4	$8	Required
C2	**3M Dual Lock or heavy-duty hook-and-loop strips** Hold the power strip, the monitor's power brick and the camera bracket in place.	1 pack	$10	Required
C3	**Adhesive cable clips and small zip ties** Keep cables flat, and away from the fan and the camera cable.	1 pack	$8	Required
C4	**Slim surge-protected power strip, 3 outlets, flat plug, 6 ft cord** Sits in the space below the monitor. The Pi supply and the monitor plug into it, so only one cord leaves the mirror. Look for widely spaced outlets.	1	$18	Required
	Subtotal		**$44**	

Mirror and light control

Two-way mirror reflects most of the light that hits it and lets some through. A bright screen behind it shows clearly; a dark area stays a mirror. Everything behind the glass that isn't screen must be black, or it shows through faintly.

MIRROR TYPE	TYPICAL COST	GOOD	WATCH OUT FOR	USE IT FOR
Two-way acrylic	$15–30 / sq ft	Light, won't shatter, easy to cut	Flexes, so reflections look wavy on larger sizes; scratches easily	Budget builds 22" and under
Two-way glass	$25–60 / sq ft	Flat, true reflection; resists scratches; any glass shop can cut it	Heavy; wear gloves, edges are sharp until seamed	**Recommended**
Smart-mirror glass	$100–150 / sq ft	Made for displays: brighter screen and a clearer mirror at once	Price; specialty suppliers only	Premium builds, bright rooms

Mirror film stuck to ordinary glass is the cheapest route, but it tends to bubble and haze. We don't recommend it.

REF	ITEM	QTY	EST.	NEED
D1	**Two-way mirror glass, 24 × 18", 4–6 mm** Cut to your size from the sizing table, with edges seamed. 6 mm (¼") is the stiffest; 4 mm saves about 4 lb. Read out the order text below to the shop. Local glass shop, or online cut-to-size: twowaymirrors.com (smart-mirror glass), TAP Plastics (two-way acrylic)	1	$120	Required
D2	**Black foam board, 20 × 30", 3/16"** The blackout mask. Cut it to the glass size, then cut a window the size of the lit screen area and a hole for the camera lens. Matte black poster board also works.	1–2	$8	Required
D3	**Black electrical or gaffer tape** Covers status lights on the monitor, Pi and camera, which would otherwise glow through the mirror at night.	1 roll	$5	Required
D4	**Glass cleaner and microfiber cloths** Clean both faces before you close the box. Dust behind the glass is hard to get at later.	1	$8	If needed
	Subtotal, required parts		**$133**	

What to ask the glass shop for

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Two-way (transparent) mirror glass, 6 mm (¼"), cut to 24" × 18".
Seamed edges, please. Mark which side is the coated side.
It's for a mirror with a screen behind it, so I need a clear view through it from the back.
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Ask which side should face the room for their glass, and mark that side with tape before you handle it. For tempered glass, order it at the final size: tempered glass can't be cut afterwards.

2 · PARTS LIST

Frame

The recommended frame is a 3½"-deep box with a mitered front frame and a plywood back. It takes a miter saw (or a hand miter box), a drill and a few clamps. Quantities below are for 24 × 18" glass. For other sizes, use the cut list formulas on page 9.

REF	ITEM	QTY	EST.	NEED
E1	**1×4 poplar board, 6 ft** The box sides. A 1×4 actually measures ¾ × 3½". Poplar paints smoothly and stays straight; pine is cheaper (budget build, ≈ $16). Use 1×6 if your monitor is deeper than 3".	2	$28	Required
E2	**1×2 poplar board, 6 ft** The front frame. It overlaps the glass by ¾" all round to hide the glass edge and the mask.	2	$14	Required
E3	**½" plywood project panel, 2 × 4 ft** The back panel. It carries the monitor, the Pi and the frame half of the cleat, so don't swap it for hardboard. Offcuts can make a wood cleat (F1b).	1	$22	Required
E4	**½" square wood strip (dowel), 4 ft** Glass stops, glued inside the box behind the glass so it stays put when the back is off.	2	$8	Required
E5	**Wood glue**	1	$6	Required
E6	**Wood screws: #8 × 1¼" (box corners) and #6 × ¾" (back panel)** About 8 of the #8 and 14 of the #6. Pocket-hole screws work too if you have a jig.	1 pack each	$10	Required
E7	**1¼" brad nails (or finish nails)** Pin the front frame and stops while the glue sets.	1 pack	$6	Required
E8	**Wood filler and sandpaper (120 and 220 grit)**	1 each	$10	Required
E9	**Primer and paint, or stain and clear coat** Satin black or charcoal looks like a cockpit panel. Paint the inside of the box and the back of the front frame matte black whatever the outside color.	1 set	$18	Required
E10	**Self-adhesive felt or rubber bumpers** Two on the bottom back corners, so the mirror hangs plumb and doesn't mark the wall.	1 pack	$4	Required
	Subtotal		**$126**	

Not building a frame?

- **Picture-framing shop:** ask for a deep "shadow box" or a frame with a build-up (riser) giving at least 3½" inside depth, and bring your glass. Expect $150–300 instead of E1–E10.
- **Off-the-shelf shadow boxes** are almost always too shallow for a monitor. Check the inside depth, not the outside depth.
- **3D printing:** printed corner brackets can join aluminum or wood rails, but you still need a back panel strong enough for the cleat.

2 · PARTS LIST

Wall mounting and power

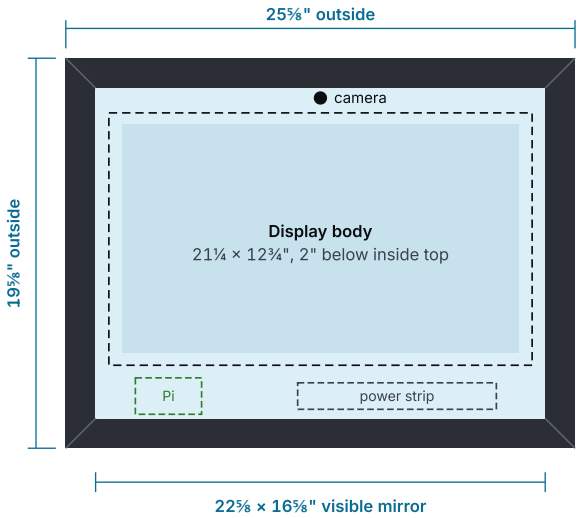
Pick one mounting method and one power option. The finished mirror is heavy and has glass in it: always screw into at least one wall stud, and use hardware rated for at least twice the finished weight.

REF	ITEM	QTY	EST.	NEED
F1	**Aluminum French cleat, 18", rated 100 lb or more** Recommended. One half screws to the wall and the other to the back panel. The mirror lifts on and off, which makes service easy, and it sits tight against the wall.	1 set	$20	Pick one
F1b	**Wood French cleat from plywood offcut** Rip a 3"-wide strip of the ½" plywood (E3) lengthwise at 45° to make both halves. Free, and just as strong.	1	$0	Pick one
F1c	**Z-bar (Z-clip) hanger set** Works like a cleat but lower profile. Good if the wall is too narrow for an 18" cleat.	1 set	$15	Pick one
F1d	**Leaning or tabletop** Stand the mirror on a dresser or desk and fit an anti-tip strap to the wall (≈ $10). Don't hang it from picture wire; the mirror is too heavy for it.	—	$10	Pick one
F2	**Wall fasteners** #10 × 3" wood screws into a stud for the wall half of the cleat. Add rated toggle bolts (snap toggles) where the cleat crosses drywall without a stud.	1 set	$8	Required
F3	**Paintable cord cover (raceway), 5 ft** Recommended. Runs the single power cord down the wall to the outlet and paints to match the wall.	1	$18	Pick one
F3b	**In-wall power kit for wall-mounted TVs** A recessed outlet behind the mirror, fed through the wall from a plug-in inlet near the floor. Buy a UL-listed kit; it installs without an electrician. Never run an ordinary power cord inside a wall.	1	$70	Pick one
F3c	**Recessed outlet installed by an electrician** The cleanest finish. Use a recessed (in-box) outlet so the plug doesn't push the mirror off the wall. Typically $150–300; not included in the totals.	—	quote	Pick one
F3d	**Visible cord** The power strip's cord hangs straight down to the outlet.	—	$0	Pick one
	Subtotal, recommended picks (F1 + F2 + F3)		**$46**	

3 · FRAME PLAN

Cut list for 24 × 18" glass

Measure the glass that actually arrives before you cut anything, and size the box from it rather than from the order.



Front view, landscape · 1×2 front frame overlaps the glass by ¾"

PART	STOCK	QTY	LENGTH
Box top + bottom	1×4	2	25⅝"
Box sides	1×4	2	18⅛"
Front frame, long	1×2, 45° miters	2	25⅝"
Front frame, short	1×2, 45° miters	2	19⅝"
Back panel	½" plywood	1	25⅝ × 19⅝"
Glass stops, long	½" square	2	24⅛"
Glass stops, short	½" square	2	17⅛"
Wood cleat (F1b)	½" plywood, 45°	2	18"

Front frame lengths are measured long point to long point.

Any other size

With glass `W × H`:

- Inside of box = `(W + ⅛") × (H + ⅛")`
- Box top + bottom = inside width + 1½"
- Box sides = inside height
- Front frame and back panel = outside size: inside + 1½" each way
- Visible mirror = inside − 1½" each way
- Long stops = inside width; short stops = inside height − 1"

Layout tips that affect what you buy

- **Place the display high.** Mount it about 2" below the inside top. That leaves room above it for the camera at eye level, and a strip about 3½" tall below it for the Pi, the power strip and the monitor's power brick.
- **Ventilate.** Drill four or five 1" holes through the top rail and the same through the bottom rail (or the back panel near the top and bottom). Heat rises out of the top and cool air comes in at the bottom.
- **Keep the controls reachable.** Cut a finger hole in the back panel over the monitor's power button, or set the monitor to switch on when power returns.
- **Camera behind the glass** works in a normally lit room. For a dim room, drill a hole just larger than the lens through the top of the front frame and mount the camera there instead.
- **Portrait builds:** turn the display 90° and switch the mirror to portrait in its display settings. Put the Pi in the space below the display.

Tools and supplies

Tools aren't included in the parts totals. Most home centers rent miter saws, or will make straight cuts for you at the lumber counter.

For every build

- Tape measure and pencil
- Combination or speed square
- Phillips and flat screwdrivers
- Utility knife and a metal straightedge (for the mask)
- Stud finder
- 4 ft or 2 ft level
- Work gloves (for glass) and safety glasses
- Computer with an SD card slot (or reader A10)
- iPhone or Android phone with the SkyGlass app

For the wooden frame

- Miter saw, or a hand saw with a miter box
- Drill/driver with bits: ⅛" pilot, countersink, 1" spade or hole saw
- Four corner clamps or two bar clamps
- Brad nailer, or a hammer and nail set
- Sanding block or random-orbit sander
- Paint brush, small roller or spray cans
- Optional: table or circular saw (for the plywood cleat), pocket-hole jig

Cost summary

GROUP	BUDGET, 22"	RECOMMENDED, 24"	PREMIUM, 27"
A · Computer and camera	$181	$186	$191
B · Display	$85	$110	$160
C · Inside the box	$44	$44	$44
D · Mirror and light control	$68	$133	$433
E · Frame	$102	$126	$126
F · Wall mounting and power	$8	$46	$98
Parts total	**≈ $490**	**≈ $645**	**≈ $1,050**

Budget: 32 GB card, 22" monitor, two-way acrylic 22 × 16", pine boards, spray paint, plywood cleat, visible cord. Premium: adds the RTC battery, a 27" 300+ nit monitor, smart-mirror glass 28 × 20" (≈ $420), and an in-wall power kit. The SkyGlass license is not included in these totals.

Shopping checklist

Recommended 24" build. Tick each part as it arrives. Write your measurements in before you order the glass.

Display body _____ × _____ depth _____ lit area _____ × _____ VESA _____

Glass to order _____ × _____ box inside _____ × _____ thickness _____

A · Computer and camera

☐	A1	Raspberry Pi 5, 4 GB	$110
☐	A2	27 W USB-C power supply	$12
☐	A3	Active Cooler	$5
☐	A4	microSD 64 GB, A2	$15
☐	A5	Camera Module 3 (or A5b webcam)	$25
☐	A6	Pi 5 camera cable, 500 mm	$3
☐	A7	Micro-HDMI to HDMI, 1 m	$8
☐	A8	M2.5 standoff kit	$8

B · Display

☐	B1	24" IPS monitor, VESA, 250+ nits	$110

C · Inside the box

☐	C1	M4 × 50 mm bolts + spacers × 4	$8
☐	C2	Dual Lock strips	$10
☐	C3	Cable clips, zip ties	$8
☐	C4	Slim 3-outlet surge strip	$18

D · Mirror and light control

☐	D1	Two-way glass, cut to size	$120
☐	D2	Black foam board	$8
☐	D3	Black electrical tape	$5

E · Frame

☐	E1	1×4 poplar, 6 ft × 2	$28
☐	E2	1×2 poplar, 6 ft × 2	$14
☐	E3	½" plywood, 2 × 4 ft	$22
☐	E4	½" square strip, 4 ft × 2	$8
☐	E5	Wood glue	$6
☐	E6	Screws #8 × 1¼", #6 × ¾"	$10
☐	E7	1¼" brad nails	$6
☐	E8	Filler, sandpaper	$10
☐	E9	Primer and paint / stain	$18
☐	E10	Felt bumpers	$4

F · Wall mounting and power

☐	F1	Aluminum French cleat, 18"	$20
☐	F2	#10 × 3" screws / toggles	$8
☐	F3	Paintable cord cover, 5 ft	$18

Also have ready

☐	–	SkyGlass license key (from your order email)
☐	–	SkyGlass app on your phone
☐	–	WiFi name and password

Questions about parts? Email support@skyglass.dev or visit skyglass.dev.